

Series No. 2: Algorithmic Formalism

Important Notes:

1. In Tutorial (TD), each algorithm must be written in algorithmic language seen in course.
2. In Practical Work (TP), translate each algorithm into C language.
3. Problem solving (analysis or algorithm) can be represented in a flowchart.

Exercise 01: Resume exercise 4 from series 1 and write the algorithm to solve each problem.

Exercise 02: For each problem, give the algorithm that solves it:

1. Write an algorithm that takes three numbers and then displays them in ascending order.
2. Write an algorithm that simulates a calculator (supporting cases of error or impossibility, e.g.: division by 0, invalid operator, etc.):
 - it requests an operation to be performed in the form of a character (exple: '+', '-', 'x', '/'),
 - it then asks for 2 numbers, performs the calculation and displays the result.
3. Water becomes ice at 0°C, it becomes steam at 100°C, and it is liquid in between. Write an algorithm that reads a temperature (in °C) and displays water status at that temperature.
4. A big store makes a discount (تخفيض) of 4% on any purchase over 15,000Da or 11% on any purchase over 49,000Da. Write an algorithm that calculates and then displays the amount to pay (المبلغ الواجب دفعه) when given the purchase amount (مبلغ الاقتناء).
5. Write an algorithm that gives the state of the aqueous product (Acid/Neutral/Base) by reading its pH.
 - indication: $0 \leq \text{pH} < 7$: acid. $\text{pH} = 7$: neutral product. $7 < \text{pH} \leq 14$: base.
6. Write an algorithm that classifies an athlete into one of the following categories by asking for their age (years):

Not Ranked	school	Minimal	Cadet	Junior	Senior
From 0 to 6	Form 7 to 13	14 and 15	16 and 17	18, 19 and 20	21 and above

7. Write an Algorithm that reads n actual values between 0 and 20 (which represent the averages of n students with $0 < n \leq 100$) and displays:
 - The percentage of students with an average < 10
 - The percentage of students with an average ≥ 10 and < 15
 - The percentage of students with an average ≥ 15
 - The algorithm must take into account incorrect entries (values < 0 and > 20). In this case, the user is asked to re-enter the value until it is correct.
8. Find all integers between 1 and n checking: $x^2 + y^2 = z^2$.
 - Examples of results: (8, 6, 10), (8, 15, 17)

9. Write an algorithm that checks whether a given character is a vowel (a,e,i,o,u,y) , a consonant (another letter), or a non-alphabetic character

➤ indication: the alphabetic letters are ordered and follow each other, 'a' < 'b' < ... < 'z').

10. Write an algorithm that determines whether a given year is leap year or not.

➤ indication: leap years are multiples of 4, but not of 100, except for millenniums who are.

11. Write an algorithm that reads a date in DDMMYY format (integer of 6 digits) and a number of days (≤ 30) to add to this date, then calculates and displays the new date in day, month and year (assume all months consisting of 30 days).

□ Example:

- date read = 271119 (day: 27, month: 11, year: 2019) and
- number of days to add = 13
- new date = day: 10, month: 12, year: 2019

➤ indication: use 3 variables for the date (one for the days, one for the months and one for the year).

12. The integer square root of a number n is calculated by bounding it by two successive additions of $(2i-1)$. The first number of additions (p) is the integer root of n. Write an algorithm that calculates the integer square root of a number.

$$\sum_{i=1}^p (2i - 1) \leq n < \sum_{i=1}^{p+1} (2i - 1)$$

➤ Example: $n=29$ and $1+3+5+7+9 \leq 29 < 1+3+5+7+9+11 \rightarrow$ integer root of 29 = 5

13. Write an algorithm that calculates the square root of a positive number according to Newton's algorithm which is based on the convergence of the sequence defined by:

$$u_0 = 1 \quad \text{et} \quad u_{n+1} = \frac{u_n + \frac{a}{u_n}}{2}$$

14. Write algorithms that calculate e^x , $\arctg(x)$, $\cos(x)$ and $\sin(x)$ according to their respective limited developments.

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$	$\arctg(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots$
$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$	$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$